EXPLAINABLE COVID-19 CHEST X-RAY CLASSIFICATION USING DENSENET121 AND GRAD-CAM

Internship Project Report

Submitted By

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Project Domain

Computer Vision and Explainable Artificial Intelligence (XAI)

Technologies Used

- Python
- TensorFlow / Keras
- DenseNet121
- OpenCV
- Grad-CAM
- NumPy
- Matplotlib
- Scikit-Learn

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Abstract

A brief summary of the project, dataset, model, explainability methods, and key results. Mention the achieved test accuracy (97.47%) and the use of Grad-CAM, Integrated Gradients, Entropy, Deletion AUC, and AOPC for explainability analysis.

1. Introduction

COVID-19 and other respiratory diseases can be identified using chest X-ray imaging. Deep learning models have shown strong performance in medical image classification; however, their lack of interpretability limits their adoption in clinical settings. This project aims to develop an explainable deep learning framework for classifying chest X-ray images into COVID-19, Normal, and Viral Pneumonia categories using DenseNet121 and Explainable AI (XAI) techniques.

Objectives

- Develop a multi-class chest X-ray classification model.
- Achieve high classification performance using transfer learning.
- Interpret model decisions using Grad-CAM and Integrated Gradients.
- Quantitatively evaluate explanation quality.

2. Methodology

Dataset

The dataset consists of chest X-ray images belonging to three classes:

- COVID-19
- Normal
- Viral Pneumonia

The dataset was divided into training, validation, and test sets.

Data Preprocessing

The following preprocessing steps were applied:

- Image resizing to 224×224 pixels
- DenseNet121 preprocessing using ImageNet normalization

- Data augmentation using ImageDataGenerator
  - Rotation
  - Horizontal Flip
  - Zoom
  - Width and Height Shift

Model Architecture

DenseNet121 pretrained on ImageNet was used as the feature extractor.

The architecture consists of:

- DenseNet121 Backbone
- Global Average Pooling Layer
- Dropout Layer
- Dense Output Layer (3 classes)

Transfer learning and fine-tuning were used to improve performance.

3. Experimental Setup and Implementation

Software Environment

- Python 3.10
- TensorFlow 2.10.1
- Keras
- OpenCV
- NumPy
- Scikit-Learn
- tf-explain
- tf-keras-vis

Training Configuration

Parameter	Value
Input Size	224x224
Optimizer	Adam

Classes	3
Loss	Categorical Crossentropy
Fine Tuning	Yes

Implementation Workflow

1. Data preprocessing and augmentation
2. DenseNet121 training
3. Model evaluation
4. Grad-CAM generation
5. Integrated Gradients generation
6. Entropy evaluation
7. Deletion AUC evaluation
8. AOPC evaluation

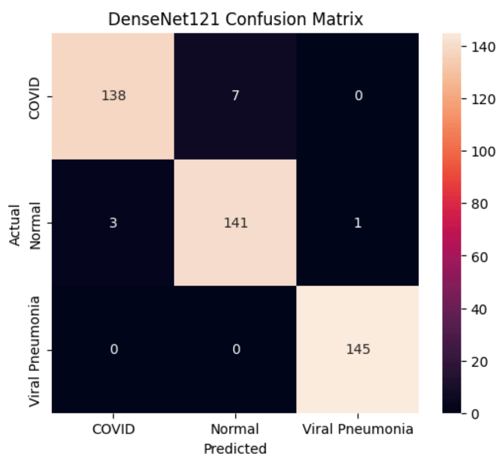
4. Results and Explainability Analysis

Classification Performance

	precision	recall	f1-score	support
COVID	0.98	0.95	0.97	145
Normal	0.95	0.97	0.96	145
Viral Pneumonia	0.99	1.00	1.00	145
accuracy			0.97	435
macro avg	0.97	0.97	0.97	435
weighted avg	0.97	0.97	0.97	435

Metric	Value
Training Accuracy	98.30%
Validation Accuracy	96.33
Test Accuracy	97.47%

Confusion Matrix



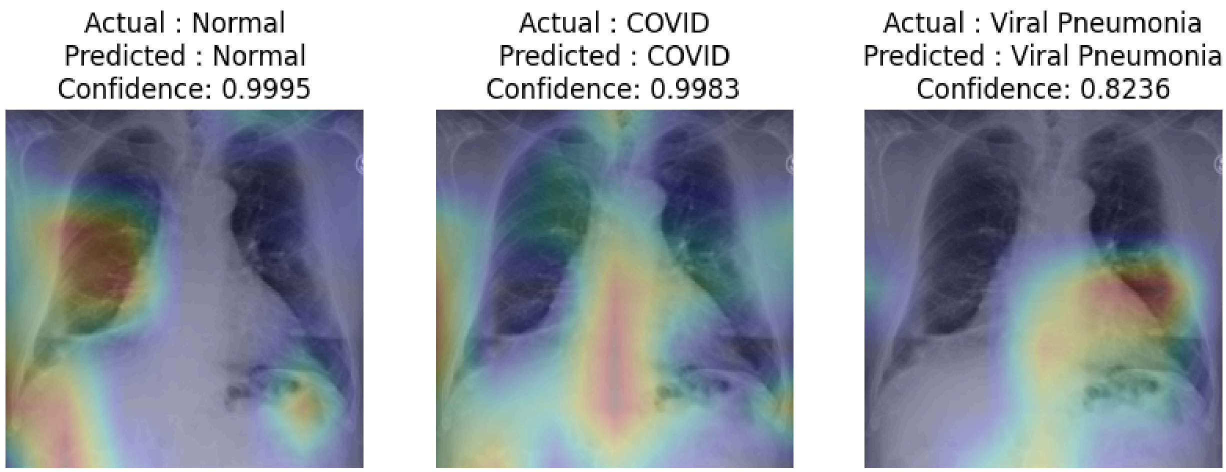
The confusion matrix indicates strong class separation with only a small number of misclassifications.

Explainability Analysis

Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to identify image regions that contributed most significantly to the model's predictions. Grad-CAM generates visual heatmaps by computing gradients of the target class with respect to the final convolutional feature maps of the network.

For each test image, Grad-CAM highlighted the regions that influenced the DenseNet121 classifier during decision-making. The generated heatmaps were overlaid on the original chest X-ray images to provide intuitive visual explanations.



Average Class-wise Heatmap Analysis

To identify common attention patterns, Grad-CAM heatmaps were averaged across multiple correctly classified samples within each class.

The average heatmaps revealed that:

- COVID-19 predictions consistently activated central and pulmonary regions.
- Normal samples exhibited more distributed activation patterns.
- Viral Pneumonia cases showed broader attention across multiple lung regions.

Misclassification Analysis

Misclassified samples were examined using Grad-CAM to better understand model failure cases. In several instances, the model focused on regions that did not strongly correspond to disease-specific patterns. Some COVID-19 images exhibited visual characteristics similar to Viral Pneumonia, leading to confusion between classes.

5. Quantitative Evaluation

Entropy

Metric	Value
Mean Entropy	10.49
Standard Deviation	0.25

Deletion AUC

Metric	Value
Mean Deletion AUC	0.4609
Standard Deviation	0.3705

AOPC

Metric	Value
Mean AOPC	0.5002

The quantitative metrics indicate that Grad-CAM provides focused and faithful explanations of model predictions.

Key Observations

- DenseNet121 achieved strong classification performance with a test accuracy of 97.47%.
- Grad-CAM consistently highlighted clinically relevant lung regions.
- High-confidence predictions produced more focused attention maps.
- Quantitative evaluation confirmed the faithfulness of generated explanations.
- Viral Pneumonia cases exhibited more diffuse attention patterns compared to COVID and Normal cases.

6. Reproducibility, Limitations and Future Work

Reproducibility

To reproduce the project:

1. Clone the GitHub repository.
2. Install dependencies using requirements.txt.
3. Organize the dataset into train, validation, and test folders.
4. Run the training notebook.
5. Execute explainability notebooks.

Assumptions

- Images are correctly labeled.
- Dataset classes are mutually exclusive.
- Transfer learning from ImageNet is suitable for chest X-ray classification.

Limitations

- Evaluation was performed on a single dataset.
- Clinical validation by radiologists was not conducted.
- Explainability methods provide approximate interpretations rather than causal explanations.

Future Work

- Evaluate additional explainability methods such as SHAP and LIME.
- Incorporate lung segmentation before classification.
- Test the framework on external datasets.

7. References

1. Huang et al., DenseNet: Densely Connected Convolutional Networks.
2. Selvaraju et al., Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization.
3. TensorFlow Documentation.
4. Keras Documentation

8. Result Links

- Accuracy Curve
- Loss Curve
- Confusion Matrix
- Classification Report
- GradCam Examples
- Misclassified GradCam Example
- Class Heatmap
- Missclassified Heatmap